SEMESTER S5

MICROCONTROLLERS AND EMBEDDED SYSTEMS

Dr. MAYADEVI N CET

Course Information

Course Code	**PCEOT501**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives

- 1 Provide a solid foundation in the principles, programming, and applications of the **8051 micro controller**
- 2. Develop expertise in the architecture and programming of ARM processors

Syllabus

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Microprocessors and Microcontrollers**-comparison of microprocessors and microcontrollers-Introduction to Embedded Systems-Application domain of embedded systems, features and characteristics, hard and soft real time systems **8051-Microcontrollers Hardware:** Microcontroller Architecture: IO Port structure, Register organization, general purpose RAM, Bit Addressable RAM, Special Function Registers (SFRs).	9
2	**Assembly programming of 8051:** Introduction to8051 assembly programming, Data types and Assembler directives, 8051 Addressing Modes, simple Assembly language programs(data transfer and arithmetic operations only) **8051 programming in C:** Data types and time delay in 8051, I/O	11

Syllabus

	programming in 8051 **8051 Timer/Counter programming in embedded C:** Programming 8051 timers, Counter programming,	
3	**8051 serial port programming in embedded C:** Basics of serial communication, 8051 connections to RS232, serial port programming in 8051. **8051 Interrupt programming in embedded C:** 8051 interrupts, external hardware and serial communication interrupt, Interrupt priority in 8051, Interrupt programming in C. **Interfacing:** LCD, ADC & DAC. Motor control: Relays and, stepper motor interfacing, DC motor interfacing and PWM using 8051	**12**
4	**Introduction to ARM processors** –ARM core-ARM Microcontroller-RISC vs CISC-Advanced features of ARM-Architecture versions-ARM Architecture-Instruction set architecture, operating modes, register set, General purpose registers- mode switching, conditional flags, Simple ALP programs on Arithmetic & logical operation, addition, subtraction, multiplication, division and factorial.	**12**

Contents

- What are embedded systems?
- Define an Embedded System
- Application Areas
- Model of an Embedded System
- Features and Characteristics
- Challenges in embedded computing system design.
- New Trends.
- Hard and Soft Real Time Systems
- Comparison of microprocessors and microcontrollers

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the general characteristics of embedded system and distinguish hard and soft real time systems	**K2**
CO2	Explain the architecture of a 8051 microcontroller	**K2**
CO3	Develop assembly language and Embedded C program for 8051 microcontroller.	**K3**
CO4	Develop assembly language for interfacing of different peripheral devices with 8051	**K3**
CO5	Explain the architecture of an ARM processor	**K3**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

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Text Books

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The 8051 Microcontroller and Embedded Systems using assembly and C	Muhammad Ali Maidu and Janice Gillespie	Pearson	2nd Edition, 2007
2	Embedded Systems: An Integrated Approach,	Lyla B Das	Pearson Education	2013
3	The 8051 Microcontroller	Kenneth J. Ayala	Thomson /Cengage Learning	3rd Edition,2007
4	Microcontroller: Architecture Assembly	Craig Steiner	Publisher: WP Publishers / Microsoft Press	
5	ARM system-on-chip architecture	Steve Furber	Addison Wesley	

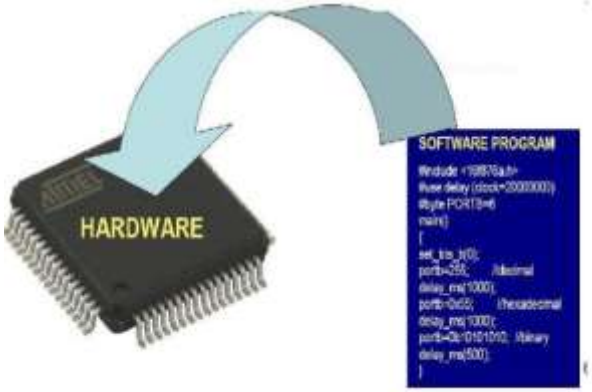
Lecture 1
Introduction to Embedded Systems

What is an embedded system?

- Embedded computing systems- Hard to define
- Nearly any computing system other than a desktop computer
- **It is an electronic system designed to perform a specific function/set of defined functions with the help of both hardware and Software components**.
- Usually **embedded as part of another system**

Definition for: Embedded System

- An Embedded System is a system that has software embedded into computer hardware, which makes a system dedicated for an application or specific part of an application or part of a larger system



- An embedded system is designed to run on its own without human intervention, and may be required to respond to events in real time.
- A special-purpose computer system
- Usually containing sensors and actuators and its control loop)

Application Domains

- **Automobiles**
- **Medicine**
- **Home Appliances**
- **Office Automation**
- **Consumer Electronics**
- **Robotics**

Application Domains

- Medical Systems
  - pace maker, patient monitoring systems, injection systems, intensive care units, …
- Office Equipment
  - printer, copier, fax, …
- Tools
  - multimeter, oscilloscope, line tester, GPS, …
- Banking
    - ATMs, statement printers, …
- Transportation
    - (Planes/Trains/[Automobiles] and Boats)
  - radar, traffic lights, signalling systems, …

Application Domains

- **Automobiles**
  - engine management, trip computer, cruise control, immobilizer, car alarm,
  - airbag, ABS, ESP, …
- **Building Systems**
  - elevator, heater, air conditioning, lighting, key card entries, locks, alarm systems, …
- **Agriculture**
  - feeding systems, milking systems, …
- **Space**
  - satellite systems, …

A “short” list of Embedded Systems

- Auto focus cameras
- ATMs
- Avionic systems
- Automatic transmission
- Battery Chargers
- Camcorders
- Cell phones
- Cordless Phones
- Digital Cameras
- DVD players
- Electronic Toys/Games
- Fax machines
- Medical Equipment
- Microwave ovens
- Modems





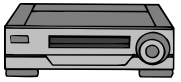
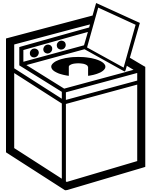





- Network cards
- Pagers
- PDAs
- Photocopiers
- Portable Video games
- Pont of Sales terminals
- Printers
- Scanners
- Satellite phones
- Teleconferencing syst
- Televisions
- Set-top boxes
- VCR’s
- Video phones
- Washers and dryers





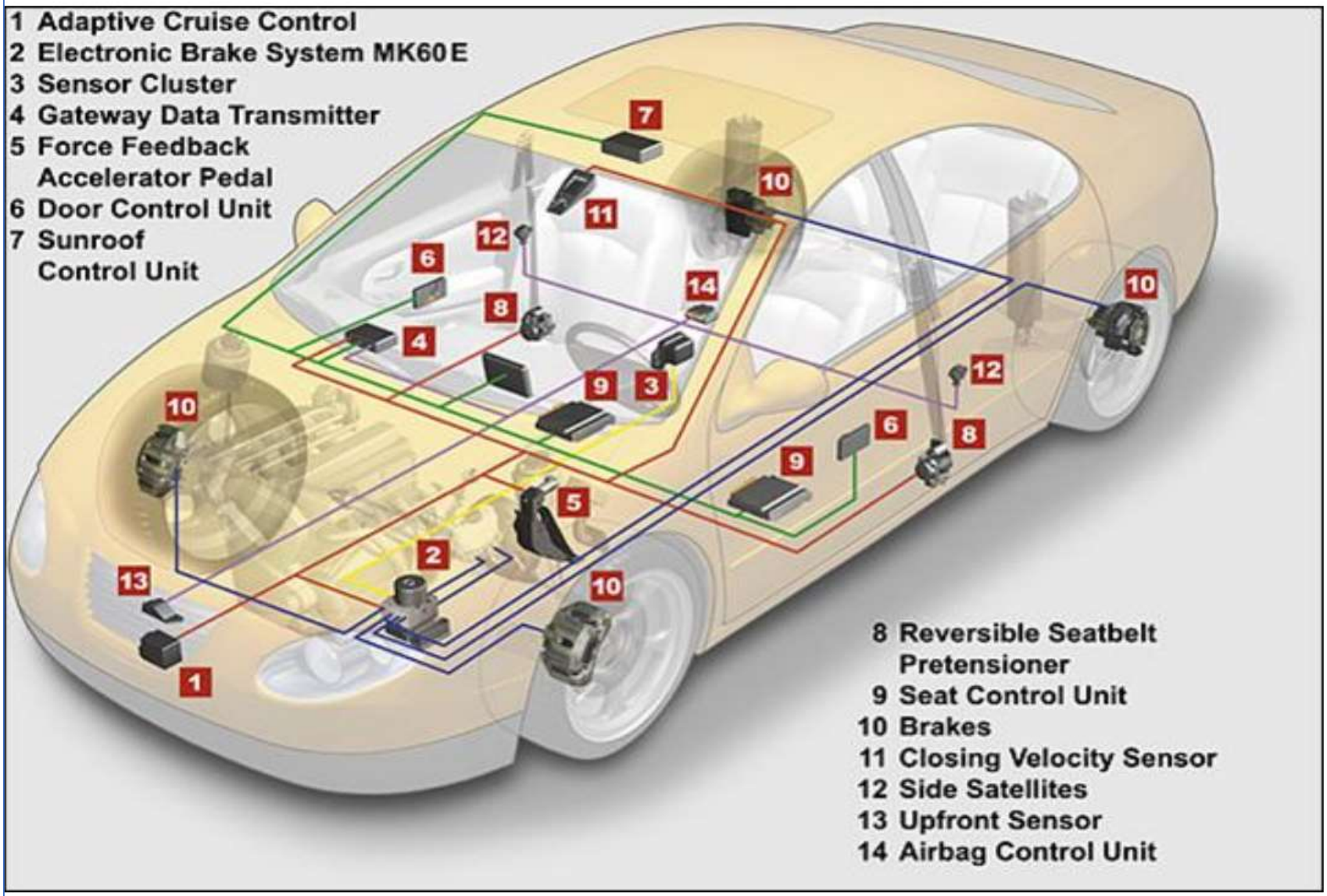



and many more ….

Applications in Automotive System

- Air Bags
- Traction Control
- The Black Box
- Automatic Parking
- Anti-lock Brake
- Adaptive Cruise
- Heads-up display Control
- Night Vision
- Drive by wire
- Satellite Radio
- Back-up collision sensor
- Telematics E.g.: OnStar
- Rain-sensing Wipers
- Navigation Systems
- Emission Control
- Tyre Pressure Monitor
- Climate Control

Application Domains



Some Common Characteristics of Embedded Systems

- Single-functioned
  - Executes a single program, repeatedly
- Tightly-constrained
  - Low cost, low power, small, fast, etc.
- Reactive and real-time
  - Continually reacts to changes in the system's environment
  - Must compute certain results in real-time without delay

Embedded System Vs General Computing System

Embedded System	General Computing System
Few applications that are known at design-time.- Speceific purpose	For executing variety of Applications.
Low power consumption is a design criteria	Low power consumption good idea but that is not guiding principle in design
Fixed amount of RAM Rom IO ports and timers etc.	Any amount of memory and peripherals can be added
Application are not alterable by the end user.	Application are alterable by the end user.
Normally part of larger system	Not part of other systems
Response time requirement critical in most of the cases	Response requirements are not normally time critical.
Must be reliable	PC needs to be reliable but since not part of critical systems occasional failures are affordable .

Design Criteria/objectives

- **Low Power dissipation –** many embedded products are battery powered so low power dissipations is a must
- **Small Code Size –** as limited ROM space is available inside chip
- **Small Physical Size-** Most devices are hand held devices or part of larger system .. So small physical size is desirable
- **High speed of Response**

Design Challenges

- **Power**
- **Price**
- **Performance**
- **Performance needs to be improved keeping power dissipation and price as low as possible**

Current Trends in ES

- **Use of Multi core processors**
- **Emergence of new embedded operating systems**
- **Newer areas of deployment of embedded systems**
- **Artificial intelligence –** The new trend in the **embedded systems industry** is the increase in the usage of Artificial Intelligence and Machine Learning
- ***Deep Learning Applications***
- ***Improved Security for Embedded Devices***
- ***Cloud Connectivity and Mesh Networking***

Real Time System

- Real-time systems have been defined as: "**those systems in which the correctness of the system depends not only on the logical result of the computation, but also on the time at which the results are produced**";
  - **Logical Correctness:** Produces correct outputs.
  - **Temporal Correctness:** Produces outputs at the right time.
    - It is not enough to say that “brakes were applied”
    - You want to be able to say “brakes were applied at the right time”

Real-Time systems

- Real-time systems often are comprised of a *controlling* system, *controlled* system and *environment*.
  - *Controlling* system: acquires information about environment using *sensors* and controls the environment with *actuators*.
  - *Real Time Systems-* ***Hard , Soft, Firm***

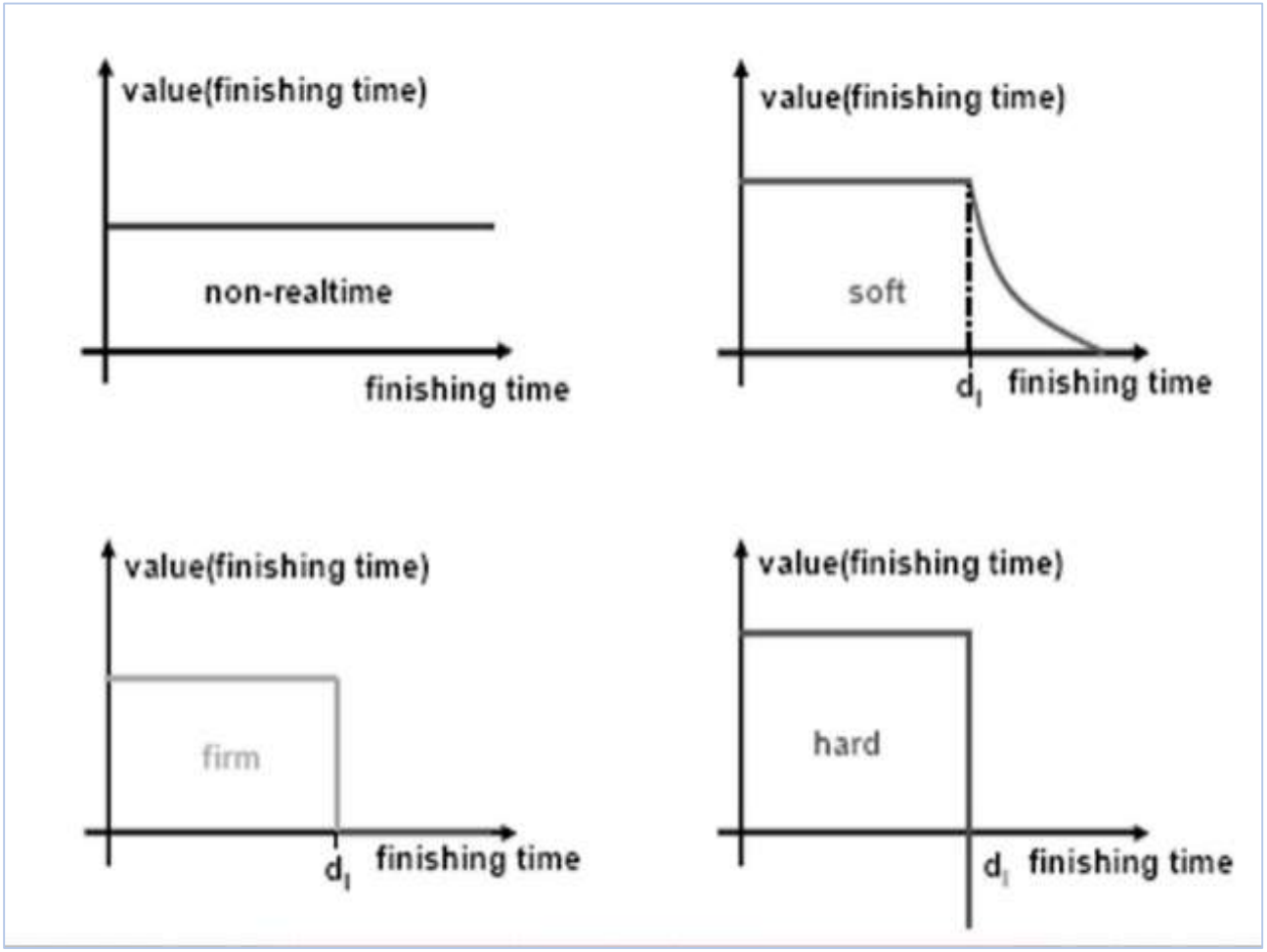
Hard Soft Real Time Systems

- The deadline of a job is the instant of time by which its execution is required to be completed
- Hard real-time — systems where it is absolutely imperative that responses occur within the required deadline. E.g. Flight control systems.
- Failure to meet constraint causes a fatal fault
- Eg:Pace maker, ABS etc

Hard Soft Real Time Systems

- **Firm real-time- dead line must be met like hard real time.. If not met no fatal effect like hard real time .. Eg.. Real time video**
- **Soft real-time** — systems where deadlines are important but which will still function correctly if deadlines are occasionally missed.
- : late completion is **undesirable but generally not fatal.**
- E.g. Data acquisition system. multimedia applications.

Real Time Systems



Lecture 2
Introduction to Microcontrollers

Microcontrollers

Dr. MAYADEVI N CET

Syllabus –Module1

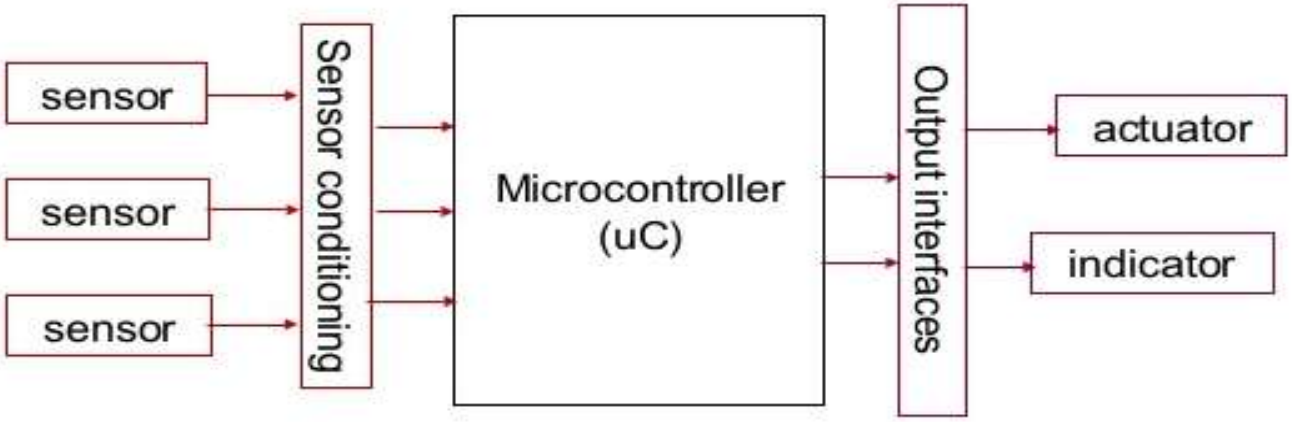
- **Introduction to Microprocessors and Microcontrollers**
  - -comparison of microprocessors and microcontrollers- Introduction to Embedded Systems Application domain of embedded systems, features and characteristics, hard and soft real time systems
- **8051-Microcontrollers Hardware:**
  - Microcontroller Architecture: IO Port structure, Register organization, general purpose RAM, Bit Addressable RAM, Special Function Registers (SFRs).

Contents

- Introduction to microcontrollers
- Microprocessor Vs Microcontroller
- 8051- Microcontrollers – Hardware
- Microcontroller Architecture and programming model
- I/O Port structure
- Register organization
- General purpose RAM
- Bit Addressable RAM
- Special Function Registers (SFRs).

Block Schematic Model of an Embedded System

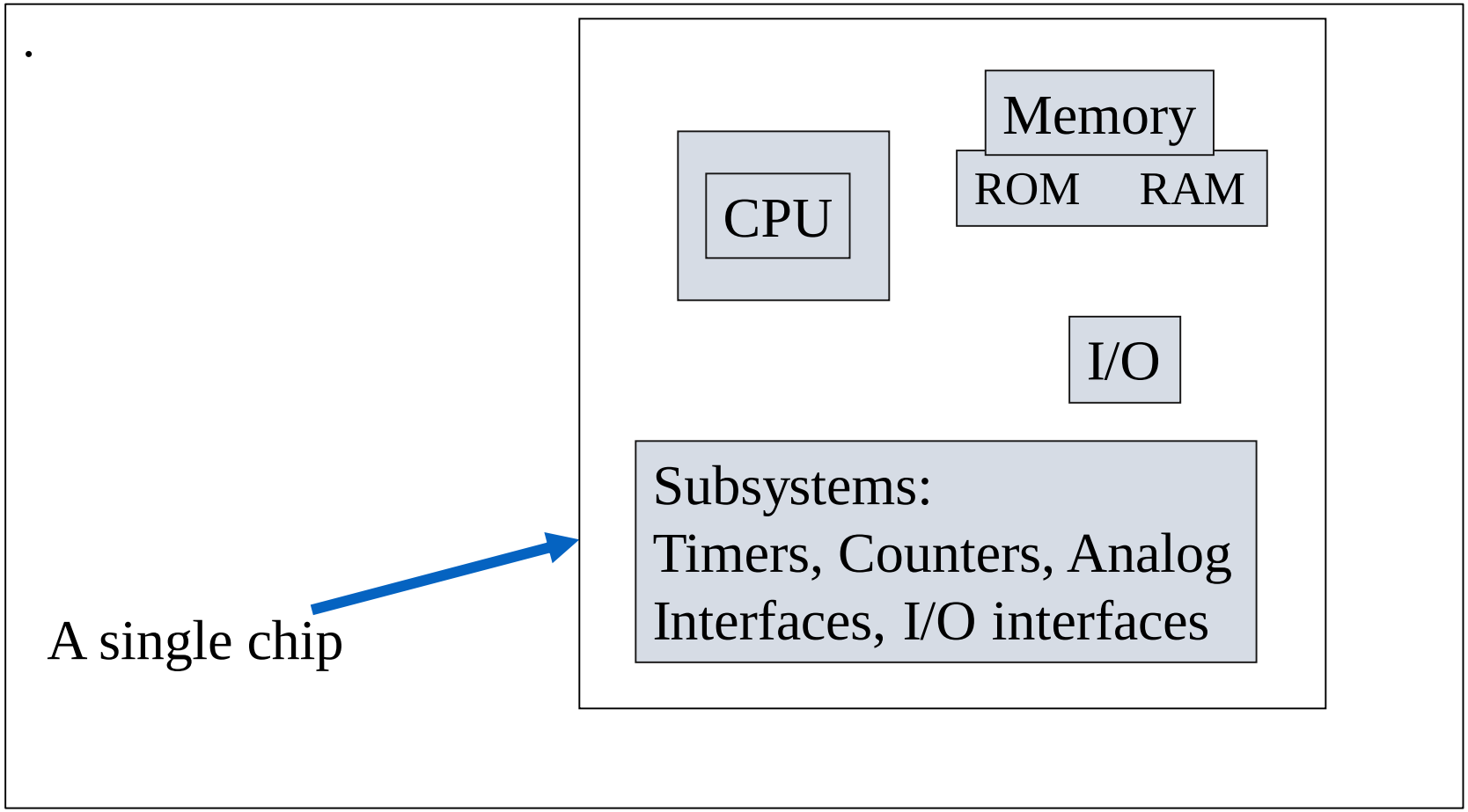
- Sensors, Microcontroller, Actuators



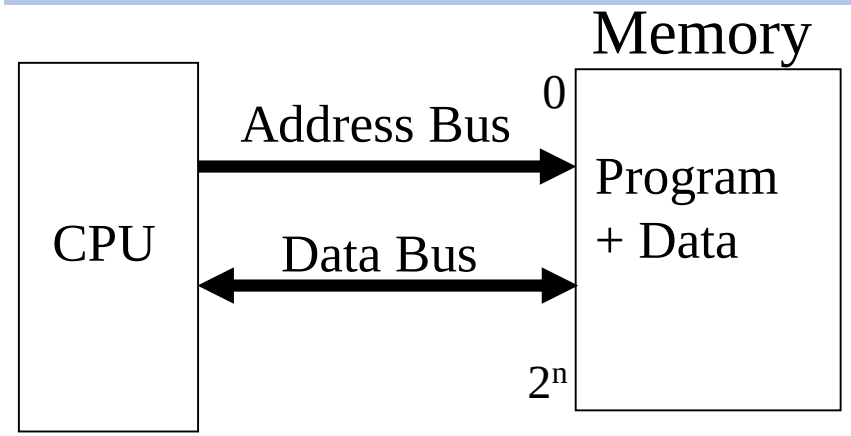
MICROCONTROLLER

- A microcontroller is a functional computer system-on-a-chip.
- It contains a processor, memory, and programmable input/output peripherals.
- Microcontrollers include an integrated CPU, memory (a small amount of RAM, program memory, or both) and peripherals capable of input and output.

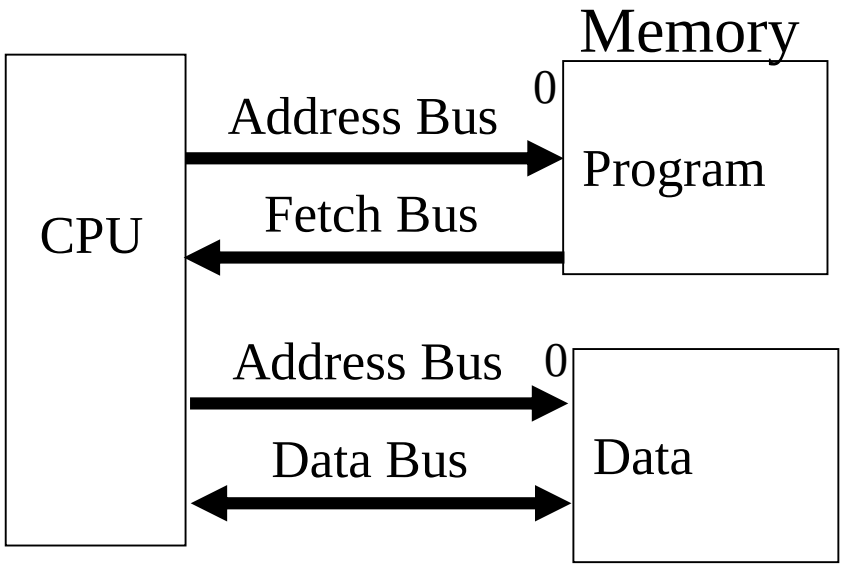
Microcontrollers



Microcontroller Architectures

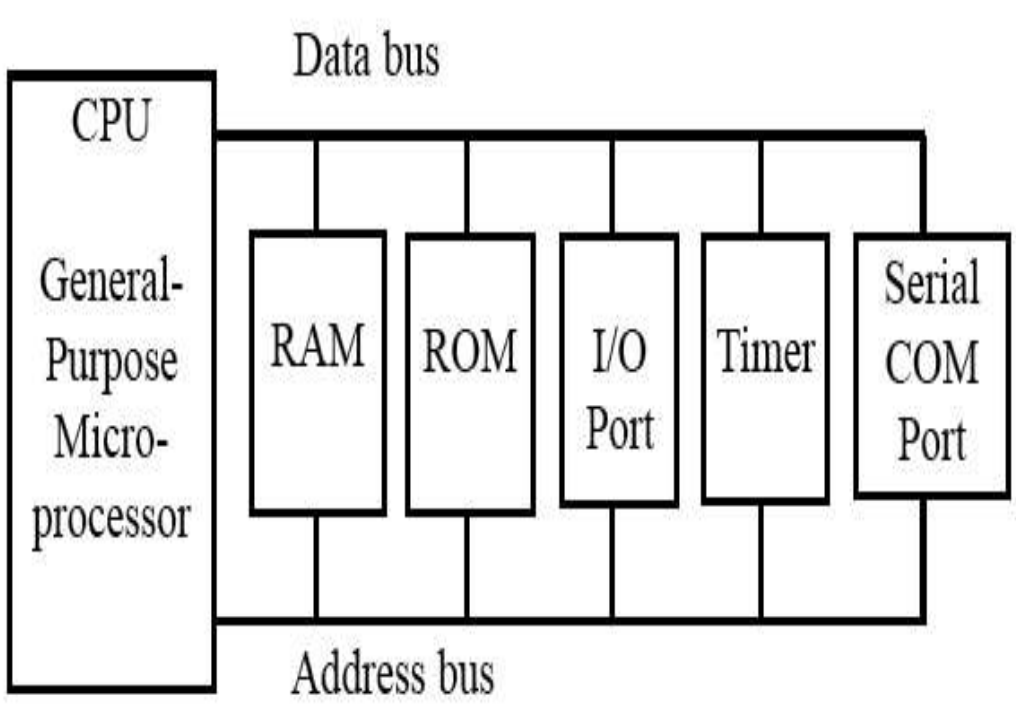


Von Neumann Architecture



Harvard Architecture

Microcontroller versus general-purpose microprocessor



(a) General-Purpose Microprocessor System



(b) Microcontroller

Micro Processor Vs Micro Controller

µP	µC
General-purpose microprocessorscontains **No RAM** **No ROM** **No I/O ports**	**Microcontroller has** CPU (microprocessor) RAM ROM I/O ports Timer ADC and other peripherals
It is dependant unit- Must add RAM ROM etc externally Make the system bulkier and much more expensive Flexible -Have the advantage of versatility on the amount of RAM, ROM, and I/O ports - flexible	It is self contained unit,..Fixed RAM ROM etc that makes them ideal for many applications in which cost and space are critical
It is designed for general purpose	It is designed for special purpose
Limited power saving options compared to µC.	Includes lots of power saving features

VARIOUS MICROCONTROLLERS

- INTEL
  - 8031,8032,8051,8052,8751,8752
- PIC
  - 8-bit PIC16, PIC18, 16-bit DSPIC33 / PIC24, PIC16C7x
- Motorola
  - MC68HC11

Common Microcontrollers

- Atmel
- ARM
- Intel
  - 8-bit
    - 8XC42
    - MCS48
    - **MCS51** ←
    - 8xC251
  - 16-bit
    - MCS96
    - MXS296
- National Semiconductor
  - COP8
- Microchip
  - 12-bit instruction PIC
  - 14-bit instruction PIC
    - PIC16F84
  - 16-bit instruction PIC
- NEC

- Motorola
  - 8-bit
    - 68HC05
    - 68HC08
    - 68HC11
  - 16-bit
    - 68HC12
    - 68HC16
  - 32-bit
    - 683xx
- Texas Instruments
  - TMS370
  - MSP430
- Zilog
  - Z8
  - Z86E02

Three criteria in Choosing a Microcontroller

1. Meeting the computing needs of the task efficiently and cost effectively
   - speed, the amount of ROM and RAM, the number of I/O ports and timers, size, packaging, power consumption
   - easy to upgrade
   - cost per unit
2. Availability of software development tools
   - assemblers, debuggers, C compilers, emulator, simulator, technical support
3. Wide availability and reliable sources of the microcontrollers.

8051

- **Intel introduced 8051, referred as MCS51, in 1981**
- **Now produced by many companies in many variations**
- **The most popular microcontroller – about 40% of market share**
- **8-bit microcontroller**

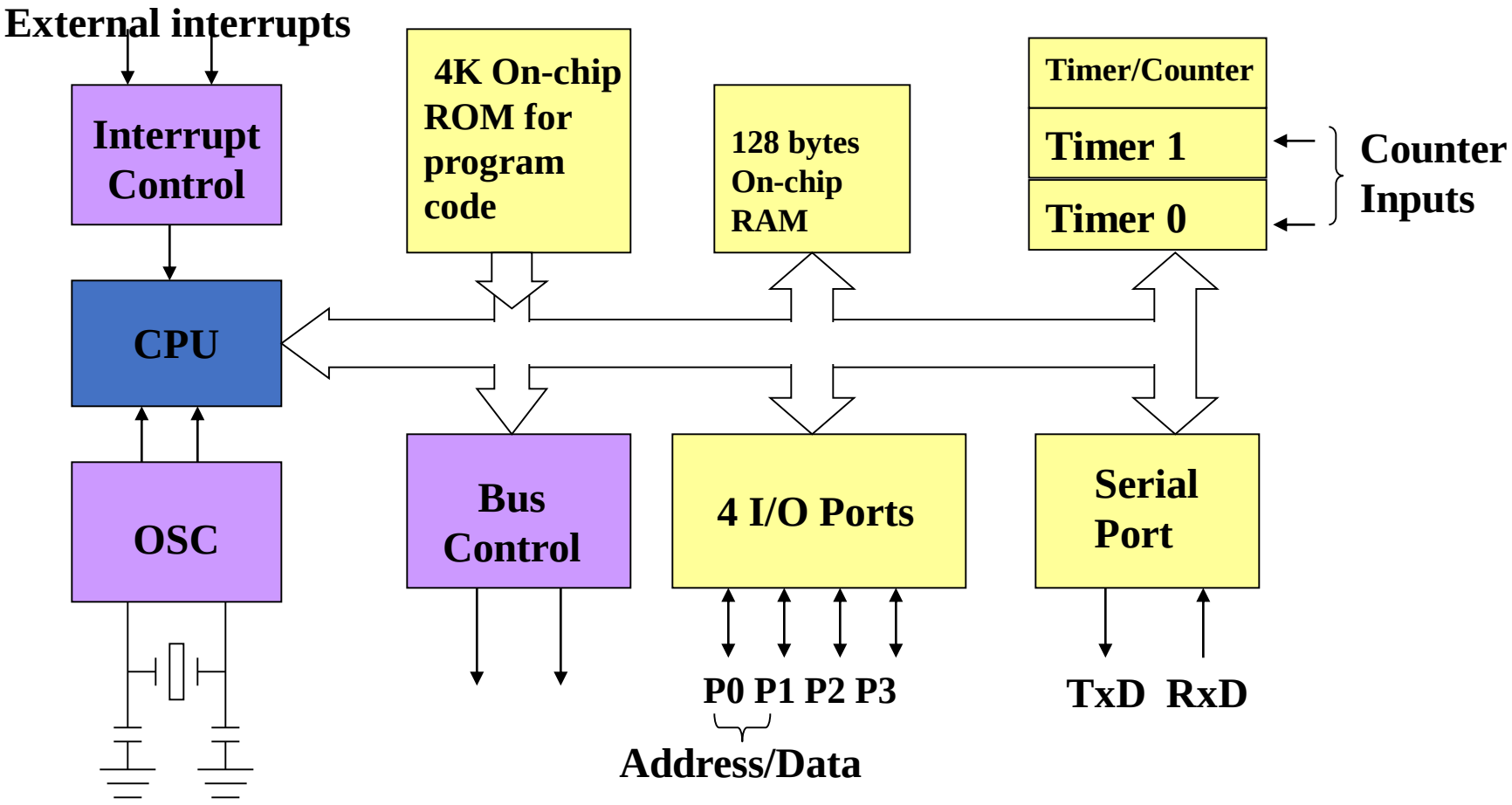
8051

- Harvard architecture (separate instruction/data memories)
- The 8051 is an 8-bit processor-The CPU can work on only 8 bits of data at a time
- **The 8051 had ?**
- **128 bytes of RAM**
- **4K bytes of on-chip ROM**
- **Two timers**
- **One serial port**
- **Four I/O ports, each 8 bits wide**
- **6 interrupt sources**

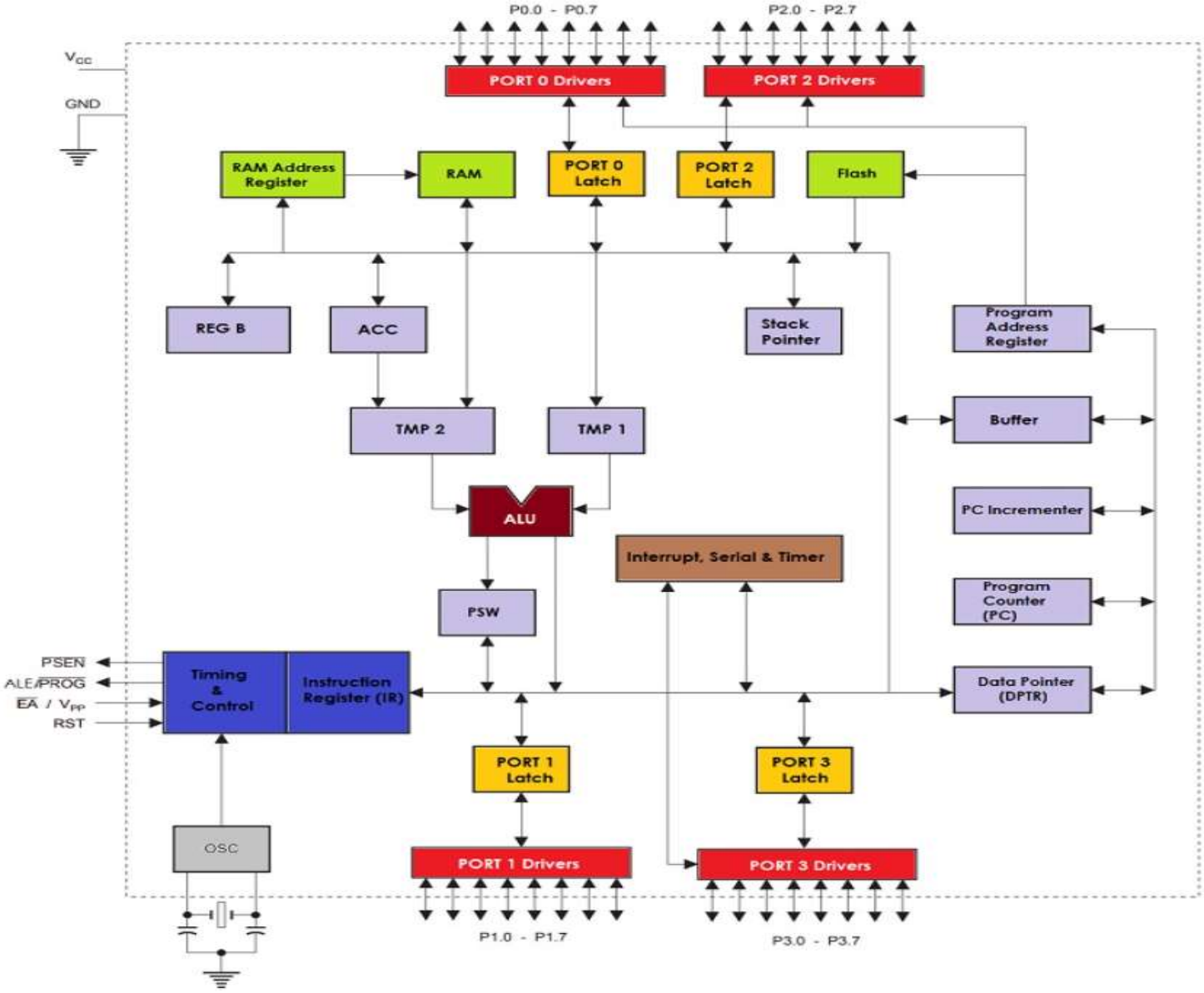
Comparison of the 8051 Family Members

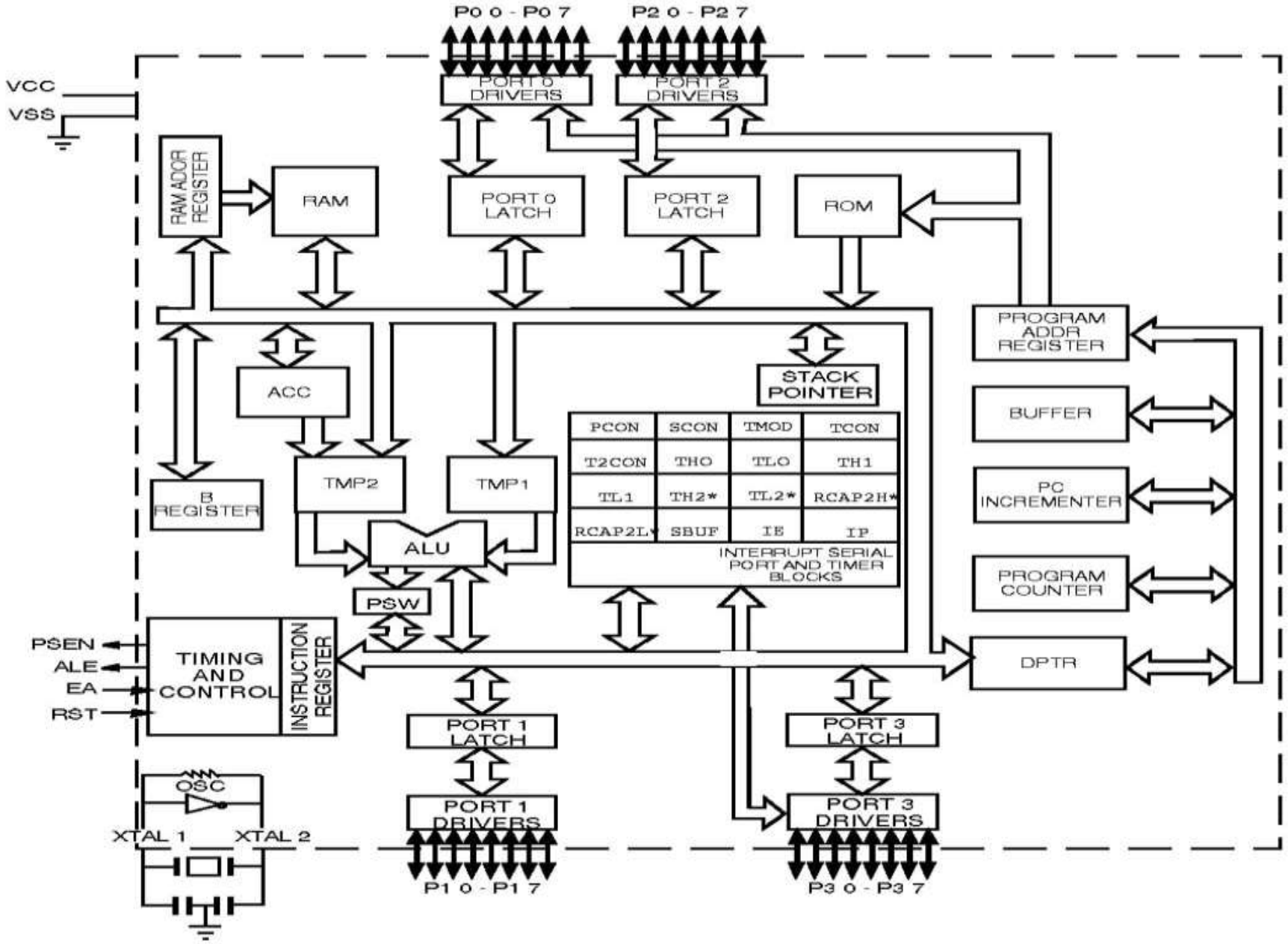
Feature	8051	8052	8031
ROM (program space in bytes)	4K	8K	0K
RAM (bytes)	128	256	128
Timers	2	3	2
I/O pins	32	32	32
Serial port	1	1	1
Interrupt sources	6	8	6

Block Diagram



Internal Architecture of 8051




P0 0 - P0 7
P2 0 - P2 7
PORT 0 DRIVERS
PORT 2 DRIVERS
VCC
VSS
RAM ADDR REGISTER
RAM
PORT 0 LATCH
PORT 2 LATCH
ROM
PROGRAM ADDR REGISTER
ACC
STACK POINTER
BUFFER
PCON
SCON
TMOD
TCON
T2CON
TH0
TL0
TH1
TL1
TH2*
TL2*
RCAP2H*
RCAP2L*
SBUF
IE
IP
INTERRUPT SERIAL PORT AND TIMER BLOCKS
B REGISTER
TMP2
TMP1
ALU
PC INCREMENTER
PSW
PROGRAM COUNTER
PSEN
ALE
EA
RST
TIMING AND CONTROL
INSTRUCTION REGISTER
DPTR
PORT 1 LATCH
PORT 3 LATCH
OSC
XTAL 1
XTAL 2
PORT 1 DRIVERS
PORT 3 DRIVERS
P1 0 - P1 7
P3 0 - P3 7

Features of 8051

- 8 bit CPU
- 16-bit program counter(PC) and data pointer (DPTR)
- 8-bit program status word(PSW)
- 8-bit stack pointer
- 4 Kbytes of on-chip program memory(ROM)
- 128 bytes of on-chip data random access memory(RAM)
- Data Bus – 8 bit
- Address Bus -16 bit

Features of 8051

- 64Kbytes of program memory address space
- 64Kbytes of data memory address space
- Two 16 bit timers/counters
- 16 bit address bus multiplexed with port 0 and port 2
- 32 bidirectional I/O lines can be either used as 8 bit ports
- On-chip clock oscillator
- Control Registers
- Serial data receiver/transmitter
- Interrupt sources

Architecture of 8051

- Processor includes
  - Arithmetic and logic unit
  - Instruction decoder
  - Timing generation unit
  - Accumulator
  - B register
  - Status register

Arithmetic and Logic Unit

- Performing computing functions
- Accumulator is 8-bit register
- Arithmetic and logical operations – one of the operands is in A register
- Result is stored in only A
- C(carry), AC(auxiliary carry), O(overflow) and P(Parity) – status register

Instruction decoder and control

- When an instruction is fetched – it is loaded in the instruction register
- Decoder decodes the instruction and establishes the sequence of events to follow
- Instruction cycle: Instruction cycle is defined as the time required for completing the execution of an instruction
- Instruction register is not programmable
- It cannot be accessed through any instruction
- Control unit generates control signals necessary for communication between the processor and peripherals

CPU Registers

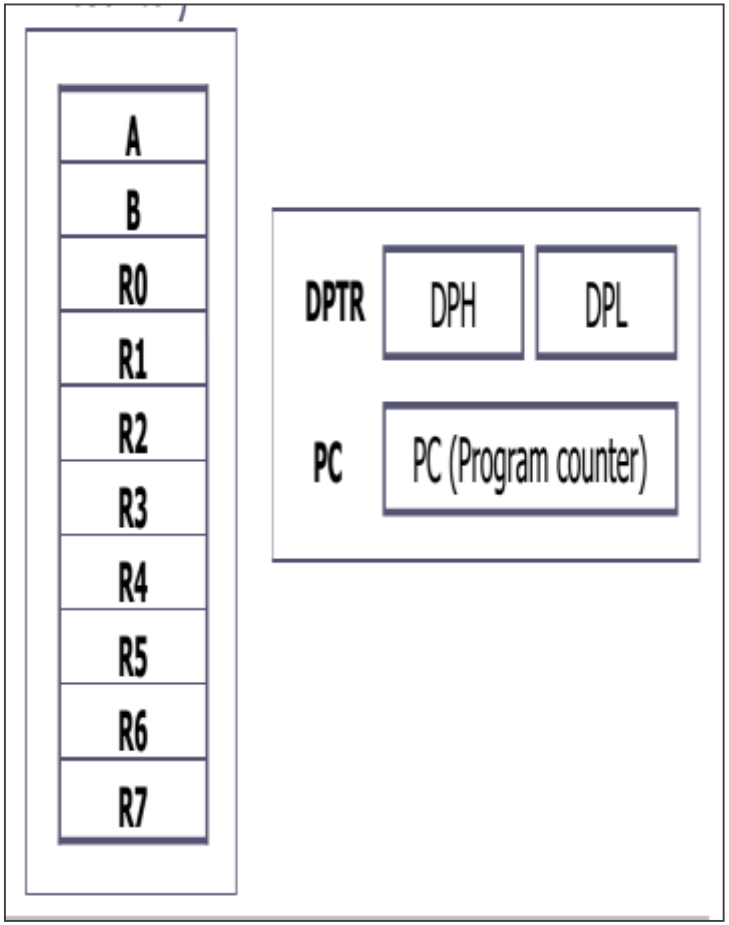
- **A Register(E0H)**
  - 8 bit
  - Direct connection with ALU
  - In multiplication- one of 8 bit operands stored in A and after operation it stores lower byte in A
  - In division- holds 8 bit dividend & quotient is stored in A
- **B Register(F0H)**
  - 8 bit
  - Used with multiply and divide operations
  - In multiplication- after operation it stores higher byte in B
  - In division- holds 8 bit divisor & reminder is stored in B

Data Pointer

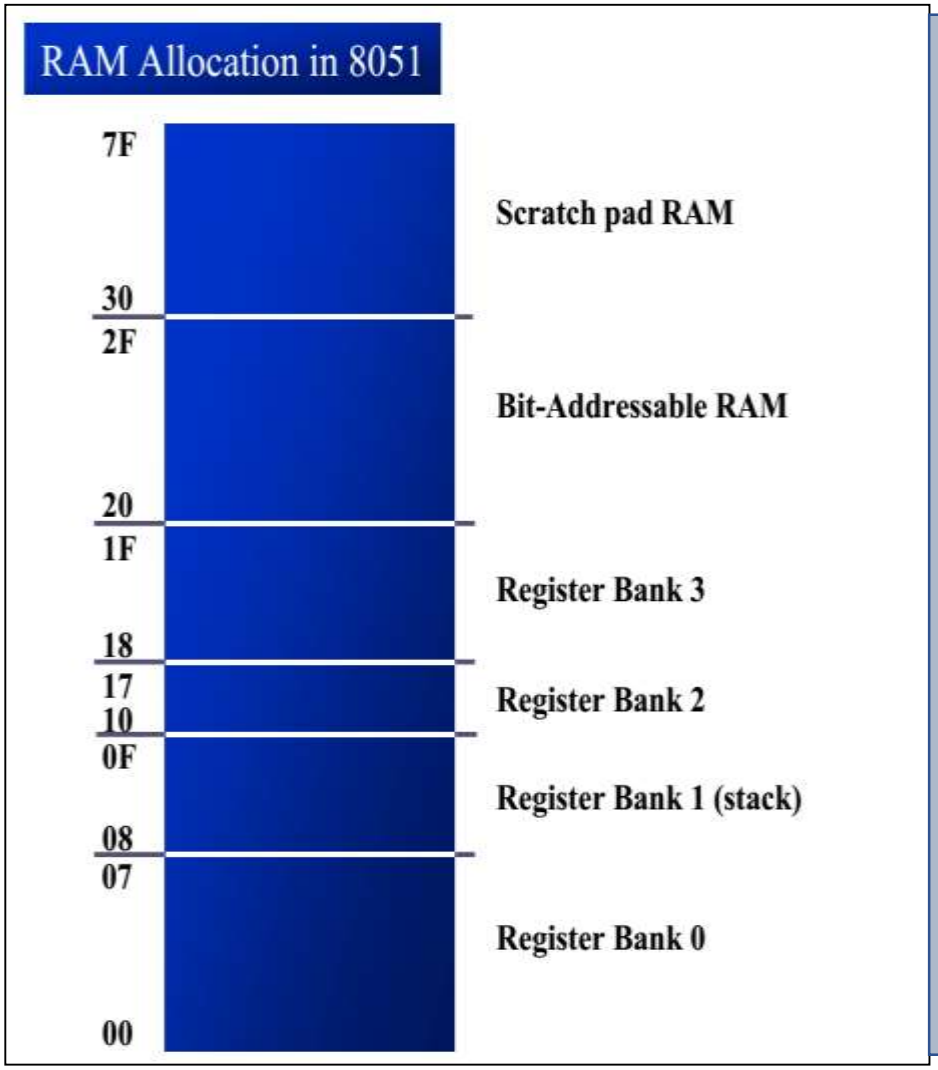
- DPTR (Data Pointer) consists of two 8 bit registers – high byte (DPH) and low byte (DPL)
- 16 bit address
- To furnish address information for internal and external program memory and for external data memory

Commonly Used registers

- The most widely used registers
- A (Accumulator) -For all arithmetic and logic instructions
- B -Used to store one of the operands for multiplication
- R0, R1, R2, R3, R4, R5, R6, R7
- DPTR (data pointer)
- PC (program counter)



RAM ARCHITECURE of 8051



There are 128 bytes of RAM in the 8051-Assigned addresses 00 to 7FH

The 128 bytes are divided into three different groups as follows:

1)A total of 32 bytes from locations 00 to 1F hex are set aside for register banks and the stack

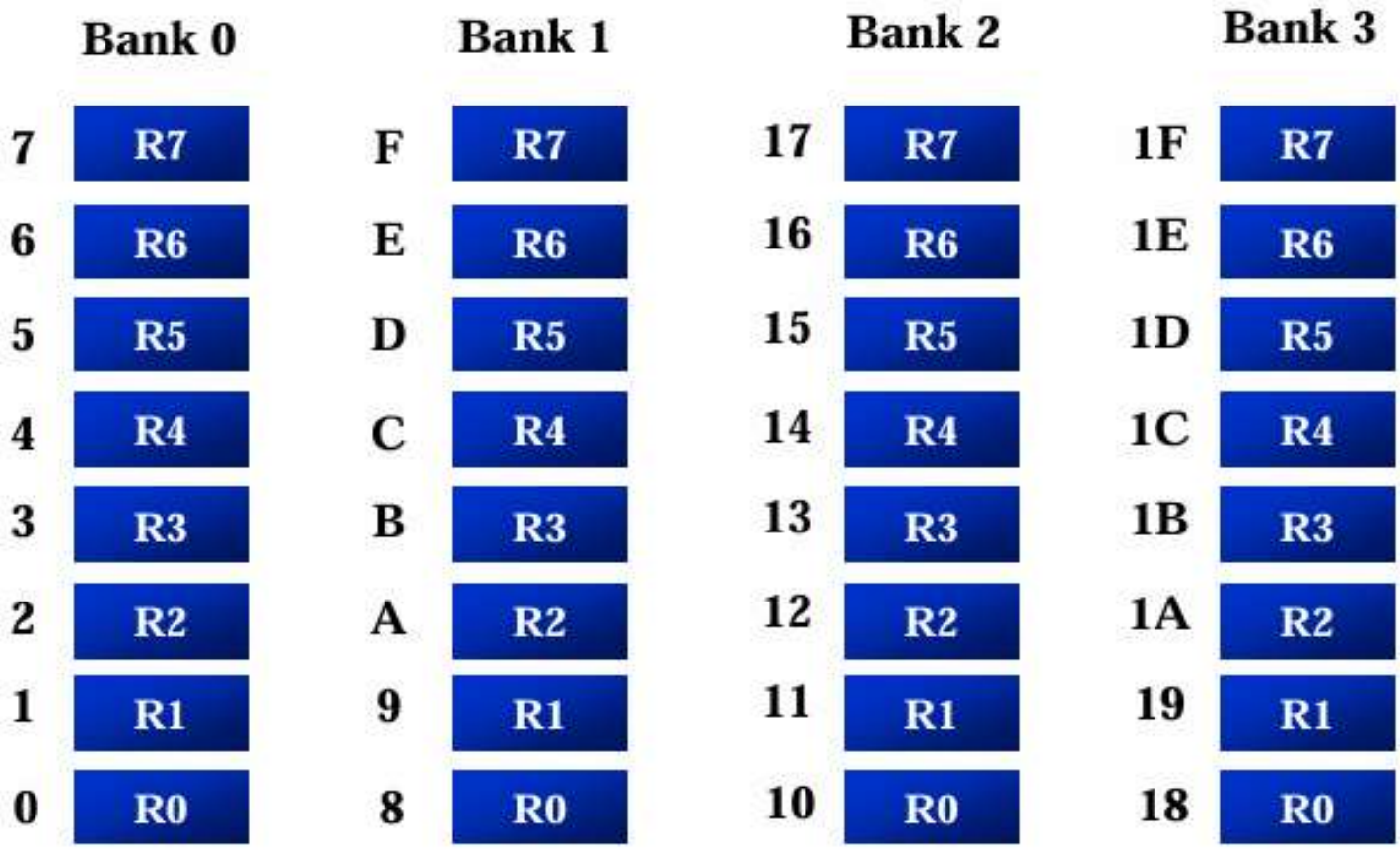
2)A total of 16 bytes from locations 20H to 2FH are set aside for bit-addressable read/write memory

3)A total of 80 bytes from locations 30H to 7FH are used for read and write storage, called scratch pad

Register Banks

- These 32 bytes register bank section is divided into 4 banks of registers in which each bank has 8 registers
- R0-R7 RAM location from 0 to 7 are set aside for **bank 0** -where R0 is RAM location 0, R1 is RAM location 1, R2 is RAM location 2, and so on, until memory location 7 which belongs to R7 of bank 0

Register banks and their RAM address



Register Banks

- Register bank 0 is the default when 8051 is powered up. We can switch to other banks by use of the PSW register **Bits D4 and D3 of the PSW** are used to select the desired register bank
- Use the bit-addressable instructions SETB and CLR to access PSW.4 and PSW.3

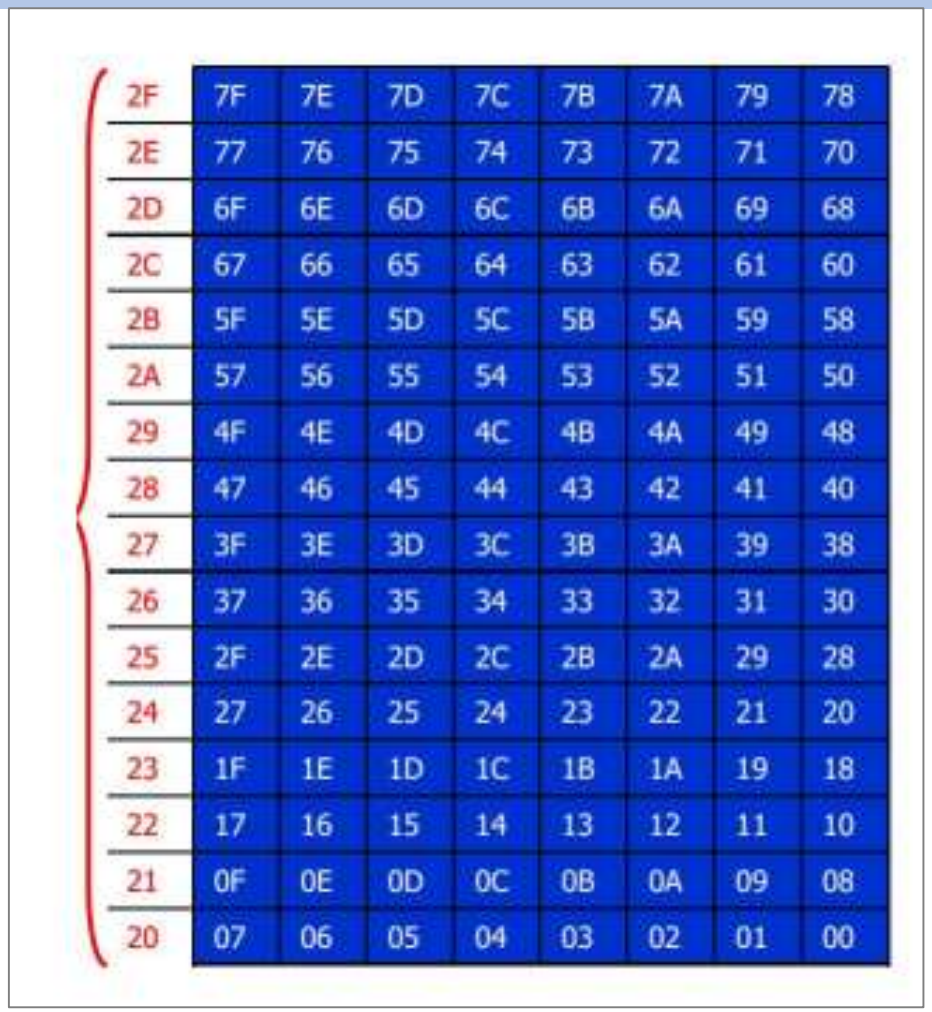
PSW bank selection

	RS1(PSW.4)	RS0(PSW.3)
Bank 0	0	0
Bank 1	0	1
Bank 2	1	0
Bank 3	1	1

Bit Addressable RAM

- The bit-addressable RAM location are 20H to 2FH.
- These 16 bytes provide 128 bits of RAM bit-addressability, since 16×8 = 128.
- 0 to 127 (in decimal) or 00 to 7FH. The first byte of internal RAM location 20H has bit address 0 to 7H

Bit Addressable RAM



2F	7F	7E	7D	7C	7B	7A	79	78
2E	77	76	75	74	73	72	71	70
2D	6F	6E	6D	6C	6B	6A	69	68
2C	67	66	65	64	63	62	61	60
2B	5F	5E	5D	5C	5B	5A	59	58
2A	57	56	55	54	53	52	51	50
29	4F	4E	4D	4C	4B	4A	49	48
28	47	46	45	44	43	42	41	40
27	3F	3E	3D	3C	3B	3A	39	38
26	37	36	35	34	33	32	31	30
25	2F	2E	2D	2C	2B	2A	29	28
24	27	26	25	24	23	22	21	20
23	1F	1E	1D	1C	1B	1A	19	18
22	17	16	15	14	13	12	11	10
21	0F	0E	0D	0C	0B	0A	09	08
20	07	06	05	04	03	02	01	00

Example 5-11

Find out to which by each of the following bits belongs. Give the address of the RAM byte in hex

(a) `SETB 42H`, (b) `CLR 67H`, (c) `CLR 0FH`

(d) `SETB 28H`, (e) `CLR 12`, (f) `SETB 05`

Solution:

(a) D2 of RAM location 28H

(b) D7 of RAM location 2CH

(c) D7 of RAM location 21H

(d) D0 of RAM location 25H

(e) D4 of RAM location 21H

(f) D5 of RAM location 20H

	D7	D6	D5	D4	D3	D2	D1	D0
2F	7F	7E	7D	7C	7B	7A	79	78
2E	77	76	75	74	73	72	71	70
2D	6F	6E	6D	6C	6B	6A	69	68
2C	67	66	65	64	63	62	61	60
2B	5F	5E	5D	5C	5B	5A	59	58
2A	57	56	55	54	53	52	51	50
29	4F	4E	4D	4C	4B	4A	49	48
28	47	46	45	44	43	42	41	40
27	3F	3E	3D	3C	3B	3A	39	38
26	37	36	35	34	33	32	31	30
25	2F	2E	2D	2C	2B	2A	29	28
24	27	26	25	24	23	22	21	20
23	1F	1E	1D	1C	1B	1A	19	18
22	17	16	15	14	13	12	11	10
21	0F	0E	0D	0C	0B	0A	09	08
20	07	06	05	04	03	02	01	00

Special Function Registers

- SFRs are registers/ memory in 8051 with specific functions.
- 8051 has 21 special function registers(SFRs) at the top of internal RAM from address 80H to FFH.
- Some SFR's are both bit addressable and byte addressable, depending on the instruction accessing the register.
- Some examples are :

SFRs of 8051

Name of the Register	*Function*	*Internal RAM Address (HEX)*
ACC	Accumulator	E0H
B	B Register (for Arithmetic)	F0H
DPH	Addressing External Memory	83H
DPL	Addressing External Memory	82H
IE	Interrupt Enable Control	A8H
IP	Interrupt Priority	B8H
P0	PORT 0 Latch	80H
P1	PORT 1 Latch	90H
P2	PORT 2 Latch	A0H
P3	PORT 3 Latch	B0H
PCON	Power Control	87H
PSW	Program Status Word	D0H
SCON	Serial Port Control	98H
SBUF	Serial Port Data Buffer	99H
SP	Stack Pointer	81H
TMOD	Timer / Counter Mode Control	89H
TCON	Timer / Counter Control	88H
TL0	Timer 0 LOW Byte	8AH
TH0	Timer 0 HIGH Byte	8CH
TL1	Timer 1 LOW Byte	8BH
TH1	Timer 1 HIGH Byte	8DH

ELECTRONICS HUB

SFR

Summary of the 8051 on-chip data memory

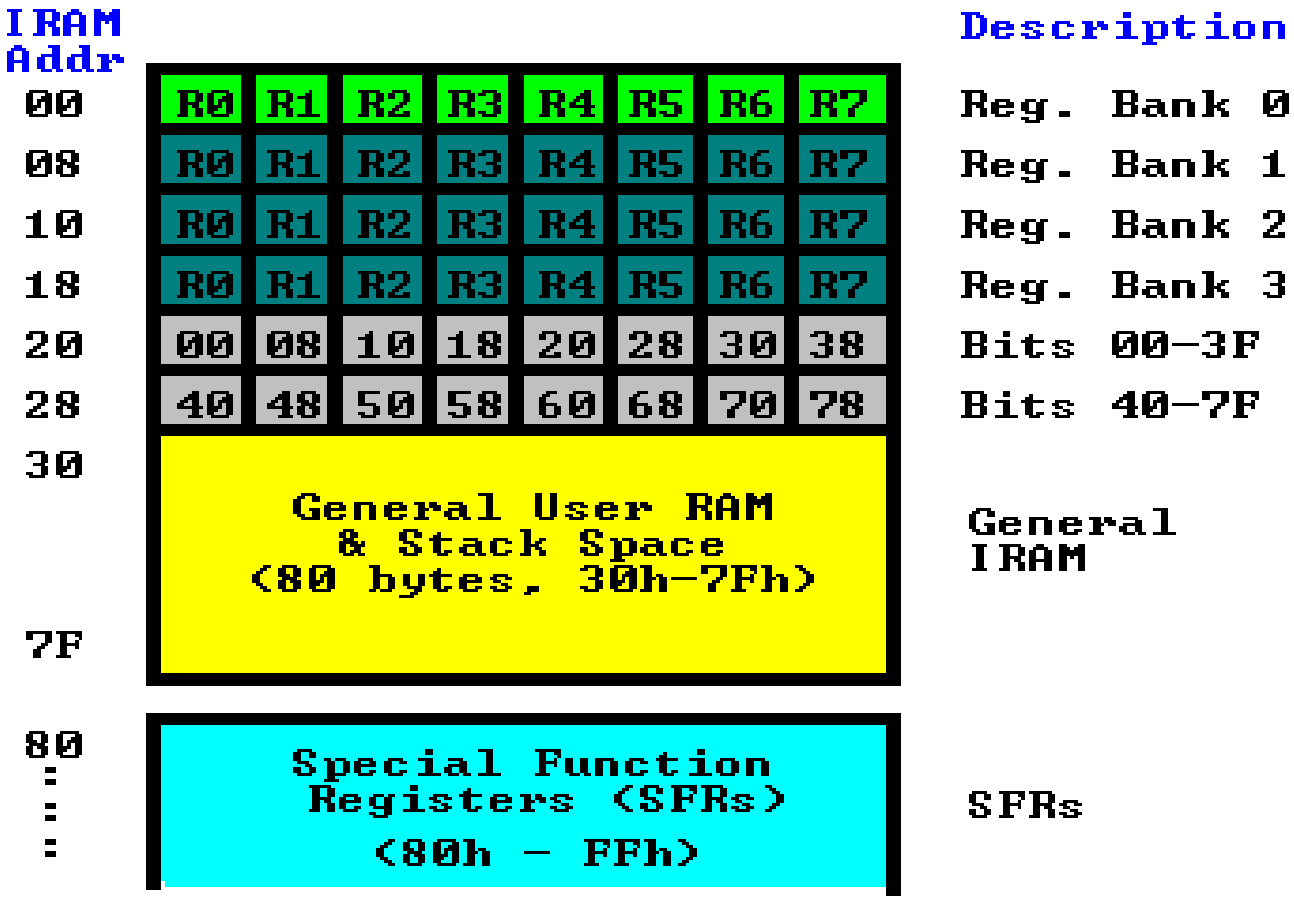
(Special Function Registers)

Byte address	Bit address									Byte address	Bit address								
98	9F	9E	9D	9C	9B	9A	99	98	SCON	FF									
										F0	F7	F6	F5	F4	F3	F2	F1	F0	B
90	97	96	95	94	93	92	91	90	P1										
										E0	E7	E6	E5	E4	E3	E2	E1	E0	ACC
8D	not bit addressable								TH1										
8C	not bit addressable								TH0	D0	D7	D6	D5	D4	D3	D2	–	D0	PSW
8B	not bit addressable								TL1										
8A	not bit addressable								TL0	B8	–	–	–	BC	BB	BA	B9	B8	IP
89	not bit addressable								TMOD										
88	8F	8E	8D	8C	8B	8A	89	88	TCON	B0	B7	B6	B5	B4	B3	B2	B1	B0	P3
87	not bit addressable								PCON										
										A8	AF	–	–	AC	AB	AA	A9	A8	IE
83	not bit addressable								DPH										
82	not bit addressable								DPL	A0	A7	A6	A5	A4	A3	A2	A1	A0	P2
81	not bit addressable								SP										
80	87	86	85	84	83	82	81	80	P0	99	not bit addressable								SBUF

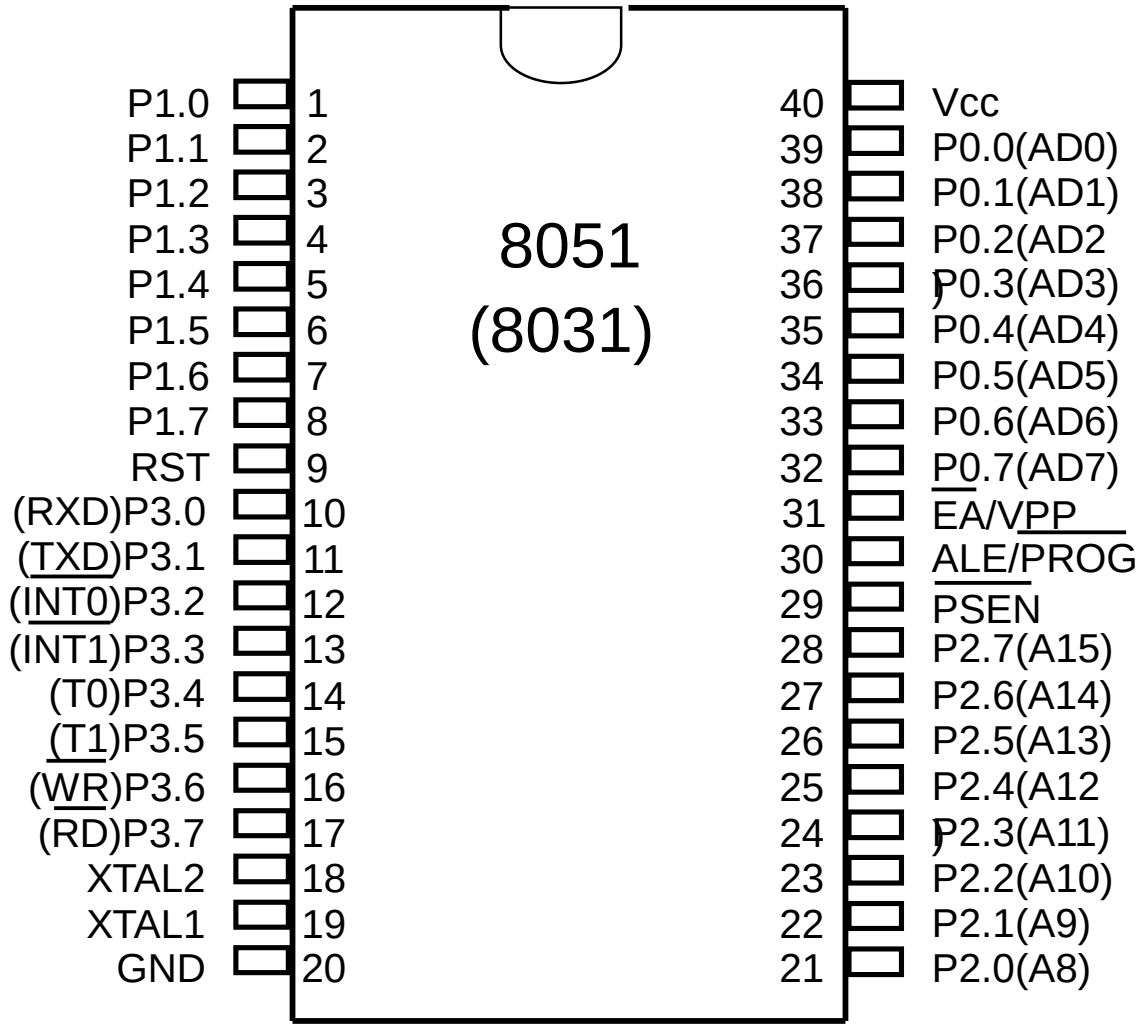
Internal RAM Organization

Special Function Registers (SFR))	**80 –FF**
General purpose registers	**30-7F**
Bit addressable space	**20-2F**
BANK 3 (R0-R7)	**18-1F**
BANK 2 (R0-R7)	**10-17**
BANK 1 (R0-R7)	**08-0F**
BANK 0 (R0-R7)	**00-07**

Memory Map


IRAM Addr
Description
00
R0 R1 R2 R3 R4 R5 R6 R7
Reg. Bank 0
08
R0 R1 R2 R3 R4 R5 R6 R7
Reg. Bank 1
10
R0 R1 R2 R3 R4 R5 R6 R7
Reg. Bank 2
18
R0 R1 R2 R3 R4 R5 R6 R7
Reg. Bank 3
20
00 08 10 18 20 28 30 38
Bits 00-3F
28
40 48 50 58 60 68 70 78
Bits 40-7F
30
General User RAM
& Stack Space
(80 bytes, 30h-7Fh)
General
IRAM
7F
80
Special Function
Registers (SFRs)
(80h - FFh)
SFRs

Pin Description of the 8051



Pins of 8051（1/4）

- Vcc（pin 40）：
  - Vcc provides supply voltage to the chip.
  - The voltage source is +5V.
- GND（pin 20）：ground
- XTAL1 and XTAL2（pins 19,18）：
  - These 2 pins provide external clock.
  - Way 1：using a quartz crystal oscillator
  - Way 2：using a TTL oscillator

.

Pins of 8051（2/4）

- RST（pin 9）：reset
  - It is an input pin and is active high（normally low）.
    - The high pulse must be high at least 2 machine cycles.
  - It is a power-on reset.
    - Upon applying a high pulse to RST, the microcontroller will reset and all values in registers will be lost.

Pins of 8051（3/4）

- /EA（pin 31）：external access
  - There is no on-chip ROM in 8031 and 8032 .
  - The /EA pin is connected to GND to indicate the code is stored externally.
  - /PSEN ＆ ALE are used for external ROM.
  - For 8051, /EA pin is connected to Vcc.
  - “/” means active low.
- /PSEN（pin 29）：program store enable
  - This is an output pin and is connected to the OE pin of the ROM.

Pins of 8051（4/4）

- ALE（pin 30）：address latch enable
  - It is an output pin and is active high.
  - 8051 port 0 provides both address and data.
  - The ALE pin is used for de-multiplexing the address and data by connecting to the G pin of the 74LS373 latch.
- I/O port pins
  - The four ports P0, P1, P2, and P3.
  - Each port uses 8 pins.
  - All I/O pins are bi-directional.

Pins of I/O Port

- The 8051 has four I/O ports
  - Port 0 （pins 32-39）：P0（P0.0～P0.7）
  - Port 1（pins 1-8）　：P1（P1.0～P1.7）
  - Port 2（pins 21-28）：P2（P2.0～P2.7）
  - Port 3（pins 10-17）：P3（P3.0～P3.7）
  - Each port has 8 pins.
    - Named P0.X （X=0,1,...,7）, P1.X, P2.X, P3.X
    - Ex：P0.0 is the bit 0（LSB）of P0
    - Ex：P0.7 is the bit 7（MSB）of P0
    - These 8 bits form a byte.
- Each port can be used as input or output (bi-direction).

